

Final Year Project Short Proposal

FYPHub: An Agentic AI-Driven Platform for Intelligent Final Year Project Management

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Introduction

FYPHub is a web-based, AI-assisted platform designed to manage the complete university Final Year Project (FYP) lifecycle in one role-aware system. It connects team formation, proposal submission, university-format compliance, originality analysis, supervisor allocation, proposal review, coordinator routing, defense scheduling, panel evaluation, progress monitoring, final archiving, and institutional knowledge reuse. The platform combines deterministic academic rules with Large Language Models (LLMs), semantic embeddings, Retrieval-Augmented Generation (RAG), GraphRAG, optimization, and risk intelligence while keeping final academic authority with authorized university personnel.

Purpose

The purpose of FYPHub is to reduce fragmented and manual FYP processes, improve proposal quality, prevent unnecessary topic duplication, make supervisor allocation more transparent, preserve review and evaluation history, and convert completed FYP records into reusable institutional knowledge. It also supports students who do not yet have a complete team through AI-based teammate matching and prevents duplicate membership in more than one active FYP group during the same cycle.

Targeted Audience

FYPHub is intended for five major roles: **Students**, **Supervisors**, **FYP Coordinators**, **Panel Members**, and **Administrators**. Each role accesses the same shared modules through role-based permissions rather than having duplicate modules for the same functionality.

Product Functions and Features

The system contains twelve integrated modules: **Admin Dashboard & Academic Management** for users, roles, academic policies, templates, deadlines, rubrics, and audit history; **Student Team & Collaboration Hub** for group creation, duplicate-member prevention, AI teammate matching, join requests, seven collaboration channels, and a group chatbot; **Student Proposal & AI Compliance Check** for short/long proposal submission, versioning, university-format validation, structural checks, and academic consistency analysis; **Proposal Similarity & Originality Check** for semantic comparison with archived FYPs, duplication evidence, contribution boundaries, and extension gaps; **Supervisor Selection & Allocation** for expertise matching, ranking, slot/capacity management, requests, and fair assignment; **Supervisor Proposal Review & Revision** for comments, editing suggestions, revision requests, two-way discussion, version comparison, and AI revision-compliance checking; **Coordinator Proposal Control** for hold, return, forward-to-panel decisions, delay detection, and version-aware academic history; **Defense & Panel Scheduling** for panel assignment, availability, date/time/room scheduling, conflict detection, and rescheduling; **Panel Evaluation & Feedback** for rubric scoring, detailed comments, AI-generated viva questions, evaluator-disagreement detection, and feedback resolution; **Project Progress & Supervision Dashboard** for supervision meetings, AI action extraction, milestones, project-health classification, risk explanation, and recovery recommendations; **Final FYP Archive & Handover** for final reports, code/data metadata, implemented-scope analysis, limitations, future work, and continuation packages; and **FYP Knowledge Search & Analytics** for natural-language institutional search, RAG/GraphRAG, multi-hop reasoning, evidence citations, and institutional FYP trends.

Social and Industrial Impact

FYPHub strengthens institutional knowledge continuity by preserving the FYP journey from team formation to final handover. It can reduce repeated project ideas, improve proposal compliance and review transparency, support fair supervisor allocation, reduce scheduling conflicts, surface project risks, and make previous FYP knowledge searchable for future batches. The project also demonstrates responsible use of AI, semantic search, optimization, analytics, and human-in-the-loop decision support in an academic management environment.

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Final Year Project (BS Artificial Intelligence)

Supervisor Consent Form

I accept the supervision of the following students with details as follows

Project Title:	FYPHub: An Agentic AI-Driven Platform for Intelligent Final Year Project Management
Student#1 Name & Registration Number	Muhammad Maaz 23108179
Student#2 Name & Registration Number	Zain Ali Kiani 23108197
Student#3 Name & Registration Number	Danish Faraz 23108160

Supervised By: Sir Hamza Imran

Signature: _____

Date: 18-09-2026